

Silicon Core Harvesting is Visible in Supercomputer Telemetry: A Fleet-Scale Study of 1,960 POWER9 Sockets

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Abstract—Processor vendors sell partially defective dies by permanently disabling faulty units, a practice known as core harvesting. The resulting per-die disable maps are proprietary and are conventionally invisible to system owners. We show that the out-of-band (BMC/IPMI) sensor telemetry routinely collected by production supercomputers exposes these maps directly, and we recover them for all 1,960 processor sockets of CINECA’s Marconi100 from the public M100 ExaData release [1]. Every socket exposes exactly 16 of 24 possible per-core thermal sensors, and the eight absent sensors always form four complete *slice* pairs $(2k, 2k+1)$ —the POWER9 unit that shares an L2/L3 block [2], [3]—with zero exceptions. We observe 442 of the 495 combinatorially possible patterns, establishing that the map is per-die rather than a fixed SKU. Using a curveball null that preserves both per-socket and per-slice marginals, we show harvested slices cluster in sensor-index space ($z = -20.8$). Thermal analysis confirms that the two cores of a slice are physically abutted (residual correlation $r = +0.51$ within-slice versus $+0.29$ for cross-slice neighbours at identical index distance), but we find only weak evidence that the slice index itself is monotonic in physical die position ($p = 0.025$; a 3×4 grid fits the thermal coupling better than a linear arrangement). We detect a sharp procurement-lot boundary at rack 22 (Welch $t = 27.3$) separating two populations with markedly different harvesting statistics. Finally, tracking every socket across the full 2.5-year, 1.67-million-socket-day record, we find harvest maps behave as near-static manufacturing constants: 89.5% of sockets never change configuration, and none of the 235 observed transitions occurs during operation—every one coincides with a reboot. Against that backdrop we isolate three episodes, on three of 1,968 sockets, in which a socket ran on 14 cores for 2–4 days, each losing exactly one slice and at least one subsequently restored: to our knowledge the first direct observation of POWER9 field core-guarding in a production system, at a measured rate of 0.15% of sockets over 2.5 years.

Index Terms—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance

I. INTRODUCTION

[TODO: write]. Framing: (i) core harvesting is standard industry practice but per-die maps are undocumented; (ii) HPC operators now collect holistic out-of-band telemetry [1], which we show carries an unintended side channel revealing manufacturing outcomes; (iii) contributions are a recovery method, the first fleet-scale characterisation of harvesting in a deployed Top-10 system, evidence of index-space clustering and of procurement-lot structure, and a thermal method for testing die-topology hypotheses.

II. BACKGROUND

A. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a 695mm^2 die in 14nm FinFET with 17 metal layers [3]. Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a slice, and each slice owns 512kB of L2 and 10MB of L3* [3], [2]. IBM’s Hot Chips 28 presentation shows the 24 cores above $12 \times 10\text{MB}$ L3 blocks, attached to a 7TB/s on-chip fabric at $256\text{GB/s} \times 12$ [2]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with 2×16 -core POWER9 at 2.6/3.1 GHz and $4 \times$ NVIDIA V100 [4], [5]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 1 shows the organisation annotated with the harvest rates we measure.

B. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, “Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23” [6]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. METHOD

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s, k] \in \{0, 1\}^{12}$ over slices $k = \lfloor \text{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \dots, 23\}$, so the index is a fixed physical-position identifier.

B. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [7], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. Row- and column-sum preservation is asserted on each of 200 draws.

C. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. RESULTS

A. Every socket is 16 cores; granularity is the slice

TABLE I
DISABLE-MAP STRUCTURE, 2020-04.

Nodes analysed	980
Sockets	1,960
Sockets reporting exactly 16 cores	1,960 (100%)
Sockets reporting 24 cores	0
Absent cores form complete $(2k, 2k+1)$ pairs	1,960 (100%)

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

B. The map is per-die, not per-SKU

442 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 7). The most common (slices 0–3) covers only 5.7% of sockets. Within a node the two sockets match in just 1.2% of cases, with mean overlap 1.648 slices against a shuffled null of 1.474 ± 0.026 ($z = +6.6$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.7% of sockets against a uniform expectation of 33.3% ($\chi^2 = 836.5$, $df = 11$, $p \ll 0.001$); Fig. 2 splits this by lot.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.243 versus 4.585 ± 0.016 ($z = -20.8$), and every significant co-harvest excess has $|\Delta k| \leq 2$ (Fig. 3): slices 0&1 ($z = +19.5$), 6&7 (+9.5), 8&9 (+8.7), 2&3 (+8.4), 4&5 (+7.9), 1&2 (+7.4), 10&11 (+7.1). All negative terms are distant pairs.

E. Slice siblings are physically abutted

Fig. 4 shows residual correlation against core index distance. At *identical* distance of 1, within-slice siblings reach $r = +0.51$ versus +0.29 for cross-slice neighbours; per-node the contrast between near (≤ 2) and far (≥ 12) pairs is $+0.45 \pm 0.20$ with 40/40 nodes positive (paired $t = +13.9$). The two cores of a slice are roughly twice as coupled as equally-numbered cores across a slice boundary, confirming both physical abutment and the slice as a physical unit.

F. The slice index is not established as the physical ordering

Free classical MDS on the pooled 24×24 matrix is dominated by noise (leading axis 20.5% of variance, correlation with core index only +0.34), so we instead score explicit candidate floorplans by correlating predicted physical distance against measured thermal coupling (Fig. 5). The identity 1×12 linear arrangement scores -0.271 , better than a random ordering ($z = -2.22$, $p = 0.025$) but *worse* than a 3×4 grid (-0.565); a hill-climbing search finds an ordering scoring -0.715 whose rank correlation with the identity ordering is only +0.34.

We therefore do not claim the sensor index is monotonic in physical die position. The index-space clustering of Fig. 3 is a robust empirical fact, but its interpretation as *spatial* defect clustering is not established by our data and must be stated as a hypothesis. **[TODO: obtain an annotated POWER9 die shot or OpenBMC sensor-ID mapping to settle this.]**

G. A procurement-lot boundary at node 440

Node IDs map to position as $\text{rack} = \lfloor \text{id}/20 \rfloor$ [6]. A changepoint scan maximises Welch t at rack 22 (node 440), $t = 27.3$ (Fig. 6).

TABLE II
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY.

Population	Sockets	$P(\text{slice } 0)$	Distinct patterns
Lot A (racks 0–21)	880	89.1%	224
Lot B (racks 22–48)	1,080	39.4%	414

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.568 vs 1.454 ± 0.008 , $z = +14.1$).

H. Configuration lifetime

POWER9 exposes mechanisms that *could* alter the active set after manufacture: predictive Dynamic Processor Deallocation, persistent GARD records read by hostboot at IPL and clearable with `opal-gard`, and the field core override used for licensing [8], [9]. Operating-system core offlining is invisible to the BMC and cannot affect sensor presence. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

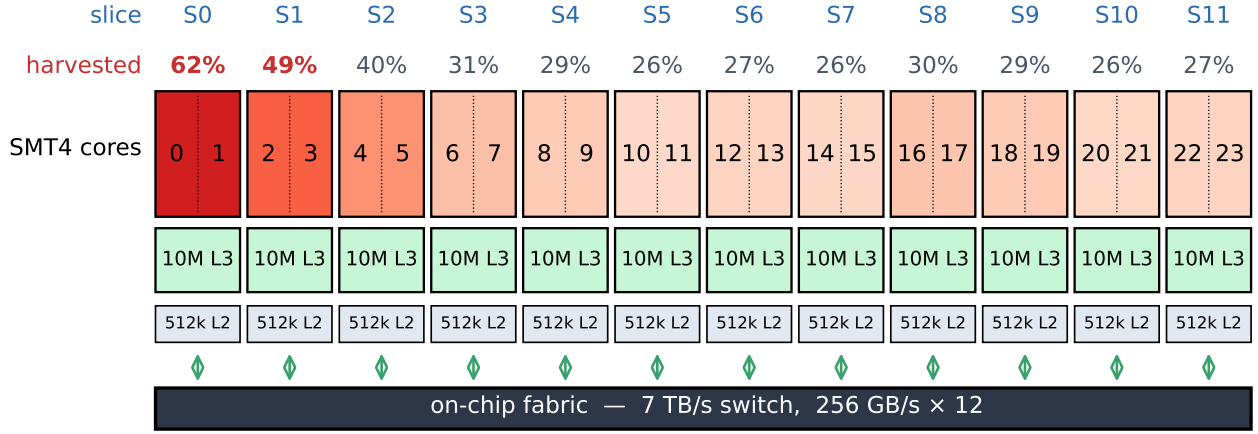


Fig. 1. POWER9 slice organisation (after [2]) annotated with the measured per-slice harvest rate across 1,960 Marconi100 sockets. Each slice holds two SMT4 cores sharing 512kB L2 and 10MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which we show is not established as the physical die ordering** (Sec. IV-F).

a) *The count is very nearly invariant:* Across all 31 months—1,671,297 socket-days over 858 days and 984 nodes—**99.985% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which *cannot* arise from slice-granular fusing and therefore proves the collector can drop individual cores; the counts above 16 are unions spanning a same-day reconfiguration. Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

b) *Configurations are near-static:* Restricting to *clean* socket-days (16 cores, each at >90% of the 4,320 samples expected at 20s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

c) *Three core-guard episodes:* The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, each with the exact signature of a field de-configuration (Table III). In every case the socket loses *exactly one slice*—a complete $(2k, 2k+1)$ pair—runs on 14 cores for 2–4 consecutive fully-sampled days, and in at least one case the slice is subsequently restored, consistent with an operator clearing the GARD record [9]. This is, to our knowledge, the first direct observation of POWER9 core guarding in a deployed system. The measured rate is **3 episodes across 1,968 sockets over 2.5 years**, i.e. 0.15% of sockets.

d) *Changes never occur during operation:* Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting (median 2d), against a background in which only 4.2% of consecutive reporting days are more than one day apart. Configuration changes are boot-time events, exactly as the guard/deconfiguration mechanism predicts.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1)

TABLE III
CORE-GUARD EPISODES. BASELINES ARE THE ADJACENT OBSERVED DAY, NOT THE SOCKET’S GLOBAL MODAL SET, SINCE TWO OF THESE SOCKETS ALSO CHANGE CONFIGURATION ELSEWHERE IN THEIR LIFE.

Socket	Dates	Days	Slice lost	Restored
347 p1	2020-03-13–16	4	4 (cores 8,9)	yes
411 p1	2021-11-13–15	3	6 (cores 12,13)	yes [†]
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	not observed

[†]restored via a day that also spans a reconfiguration.

and 342 (p0) appear to change within a single day, and produce the dataset’s only 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting gap*: on node 315 the outgoing cores {4, 5, 20, 21} report until 08:42 and the incoming cores {8, 9, 10, 11} resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, indicating a coordinated maintenance reboot. Day-granularity aggregation is therefore *conservative* here: it cannot see sub-day downtime, so our transition-gap statistics understate how consistently changes coincide with reboots.

e) *Changes concentrate on a few dates:* The 235 transitions involve 132 nodes on 81 distinct dates, but are heavily clustered: 2020-03-20 alone accounts for 63 and the top three dates for 41%. Such synchronised, fleet-wide changes are more consistent with an administrative or firmware event than with independent per-node hardware work (see OI-2).

f) *Some apparent changes are socket relabelling:* When node n ’s socket p0 loses exactly the slices p1 gains and vice versa, the dies are unchanged and only the p0/p1 labels have swapped. This accounts for **20%** of the 230 transitions where both sockets are observable.

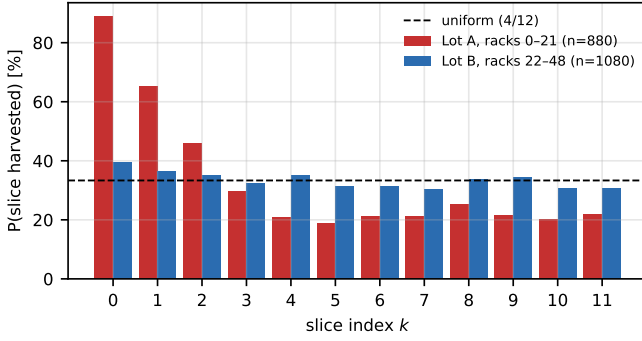


Fig. 2. Per-slice harvest probability by procurement lot.

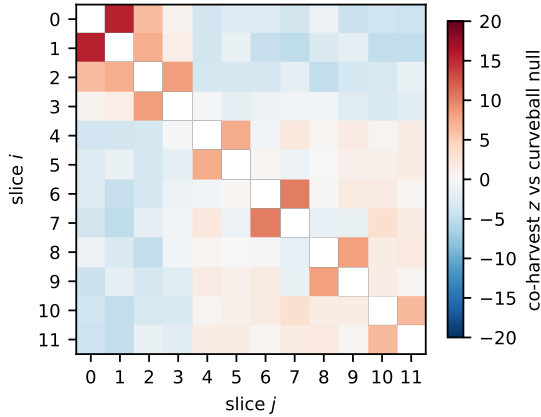


Fig. 3. Co-harvest z against the curveball null. Excess is confined to near-diagonal (index-adjacent) slice pairs.

g) *Caveat*: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. The tight clustering of 86 transitions into three days is consistent with (c) and we cannot separate these from telemetry alone. **[TODO: seek CINECA maintenance logs for March 2020.]**

V. RELATED WORK

[TODO: expand; the citations below are located but unverified.] Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [10], turbo-boost variation across nominally identical processors [11], and accelerator-level variability in GPU-rich systems [12]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [13] and spatial wafer-map models [14] describe clustered killer defects but are validated on fab data unavailable to system owners. To our knowledge no prior work recovers per-die harvesting maps from deployed-system telemetry, which is the gap this paper addresses.

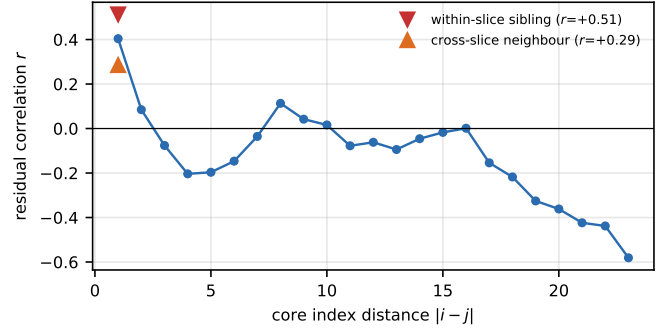


Fig. 4. Residual thermal correlation versus core index distance. The within-versus cross-slice contrast at distance 1 controls exactly for index gap.

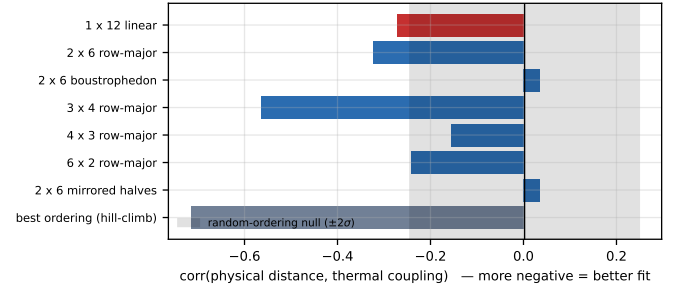


Fig. 5. Candidate floorplans scored against measured thermal coupling. The linear index ordering (red) beats a random ordering only marginally and is outperformed by a 3×4 grid.

VI. THREATS TO VALIDITY AND OPEN ISSUES

Every unresolved item is tracked with an identifier in `OPEN_ISSUES.md` in the artefact repository, which records what is known, what would settle it, and where to look. **The items marked *blocking* below must be resolved before this paper is submitted.**

- 1) **Sensor presence $\neq$ core existence.** Mitigated by 100% consistency across 1,960 sockets, agreement with the published 16-core SKU [4], and the perfect slice-pair structure. Further supported by Sec. IV-H: sub-16 counts are collection dropout.
- 2) **[OI-1, *blocking*] The sensor index may not be the physical die position** (Sec. IV-F). The spatial reading of the clustering in Sec. IV-D is a hypothesis, not a result. Needs an annotated die shot or the hostboot sensor-ID mapping.
- 3) **[OI-2, *blocking*] A March 2020 BMC firmware update could have remapped sensor IDs**, mimicking reconfiguration; 86 of 114 transitions fall in a three-day window. Needs CINECA maintenance logs. Does not affect the steady-state or boot-time-only conclusions.
- 4) **[OI-4, *blocking*] Five references are unverified** and a novelty check against prior telemetry-based binning inference is outstanding.
- 5) **[OI-3] Defect-driven versus policy-driven harvesting** cannot be separated from one machine; both are re-

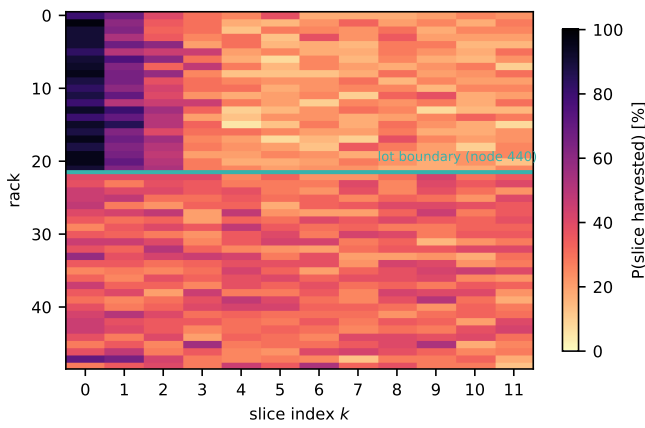


Fig. 6. Harvest probability by rack and slice. The lot boundary at rack 22 is visible as an abrupt change.

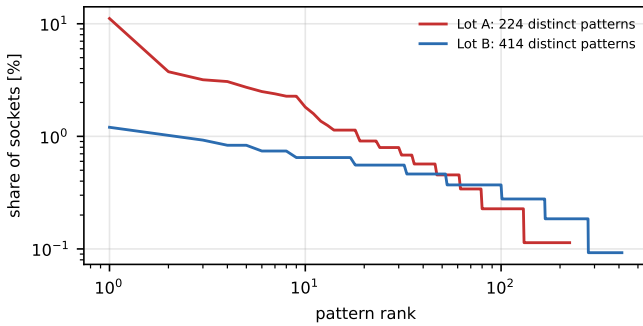


Fig. 7. Rank–frequency of disable patterns. Lot B is markedly more diverse.

ported.

- 6) **[OI-5] Collection dropout may not be random.** Monthly coverage ranges from 18% to 99.4%; if dropout correlates with lot, Sec. IV-G needs a robustness pass.
- 7) **[OI-6] One small anomaly remains unexplained:** 48 socket-days reporting a core outside the socket’s modal set. (OI-7, two nodes appearing to change within a day, is resolved: both are same-day reboots, see Sec. IV-H.)
- 8) **[OI-8] The mechanism behind p0/p1 relabelling is unknown;** any per-socket longitudinal study on this dataset risks silently mixing two dies.
- 9) **Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts; the pooled estimator may be improvable [OI-11].
- 10) **[OI-10] Single system;** findings may reflect CINECA procurement rather than POWER9 generally.

VII. CONCLUSION

[TODO: write.]

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